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Seasonal Forecasting of Ferry Passenger Demand for Operational Planning: Evidence from Bakauheni Port, Indonesia

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Abstract

Forecasting ferry passenger numbers is essential for efficient port operations and resource planning. This study applies the Seasonal Autoregressive Integrated Moving Average (SARIMA) model to forecast monthly passenger volumes at Bakauheni Port, Lampung. The SARIMA $(2,1,1)(0,1,1)_{12}$ model was selected for its ability to capture trend and seasonal patterns effectively. Diagnostic checks confirmed the model's adequacy, and validation yielded a MAPE of 11.47 percent, indicating 88.53 percent accuracy. These results show that the SARIMA model offers reliable predictive performance and can support data-driven decisions in scheduling, resource allocation, and service optimization. These results demonstrate that the developed SARIMA model possesses reliable predictive performance and can serve as a practical tool for supporting operational decision-making. The model can help this port authorities and managers optimize service provision, allocate resources more efficiently, and respond proactively to anticipated changes in passenger volume, thereby improving overall port performance and customer satisfaction in the future. Although it does not incorporate external factors, the model provides a solid foundation for future improvements and research.

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INTRODUCTION

Transportation supports community mobility and social and economic activities through the movement of individuals and goods between regions [1]. Lampung has high accessibility, with a port connecting the islands of Sumatra and Java. Every day, many people use the ferry at this port to cross between islands [2]. Based on data from the Lampung Central Statistics Agency (2023), ferries represent the dominant mode of passenger transportation, accounting for 52.32% of total passenger movements in April 2023 [3]. There are certain times when the number of passengers increases sharply, such as during school holidays or Eid al-Fitr holidays, and decreases on weekdays [4], [5]. The increase in passenger numbers during Eid al-Fitr is due to the desire of nomads to return to their hometowns [6], [7]. The recurring pattern of increases and decreases during certain periods shows that passenger traffic at Bakauheni Port has consistent seasonal characteristics from year to year.

This recurring trend reflects consistent fluctuations in passenger numbers at certain

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times of the year, causing hours of traffic congestion and passengers spilling over into the port waiting room [8]. At the same time, port capacity is limited and cramped [9]. Maritime transport operators, therefore, require reliable passenger demand estimates to anticipate fluctuations in travel volume and support informed policy decisions. Such estimates are essential for determining appropriate vessel numbers and capacity allocations to meet passenger demand efficiently [10]. In addition, these estimates are also important for planning and preparing the necessary port facilities to ensure smooth operations and optimal service.

Accurate estimates enable port authorities to make informed decisions regarding fleet management and infrastructure development, which ultimately improves the overall operational efficiency and responsiveness of the port [11], [12]. Nasution et al. [13] and Prabhadika et al. [14] emphasizes that demand forecasting plays a critical role in reducing planning errors related to vessel availability and capacity. Consequently, ports require reliable estimates of future passenger volumes to maintain service quality, particularly through data-driven forecasting approaches that support operational planning, resource allocation, and strategic decision-making in ferry transportation management at Bakauheni Port.

Data-based forecasting approaches utilize data from previous events to predict future events [15], [16]. Historical patterns formed in time series data are used to extract trend tendencies and periodic variations that appear systematically. Ferry passenger data at Bakauheni Port shows consistent seasonal fluctuations, making time series modeling such as Seasonal Autoregressive Integrated Moving Average (SARIMA) suitable for use in this case.

Applications of ARIMA and SARIMA approaches for time series forecasting across various domains are emerging, such as those of Multiningsih et al. [8], for instance employed a SARIMA model to forecast ship passenger volumes at Semayang Port in Balikpapan, highlighting its effectiveness in capturing seasonal demand patterns. Similarly, research conducted at Yos Soedarso Port in Ambon identified an optimal SARIMA specification for predicting passenger departures, achieving a Mean Squared Error (MSE) of 26.44 and a Mean Absolute Percentage Error (MAPE) of 10.23% [17]. Studies by Reza et al. [18] and Saputro et al. [19] on railway passenger forecasting in Java further confirmed the suitability of SARIMA, with the selected model SARIMA $(2,1,0)(0,1,2)_{12}$, yielding an AIC value of 2379.265. SARIMA to electricity consumption forecasting using IBM SPSS and reported comparable performance between the Box–Jenkins approach and the Expert Modeler, with an MAPE of 8.4% [20]. Collectively, these findings indicate that SARIMA, as an extension of ARIMA tailored for seasonal time series, is highly suitable for transportation demand forecasting. Its effectiveness has also been demonstrated in other application areas, including weather forecasting, due to its capability to model recurring seasonal patterns and generate reliable future predictions [21] - [25]. The SARIMA method is suitable for use in this case because it can capture regularly recurring seasonal patterns.

Several previous studies have successfully applied the Seasonal Autoregressive Integrated Moving Average (SARIMA) method to forecast passenger numbers at various ports. For example, research conducted at Semayang Port in Balikpapan [8] and Yos Soedarso Port in Ambon showed that SARIMA was able to model seasonal fluctuations and produce accurate estimates of ship passenger movements [17]. These studies highlight the effectiveness of SARIMA in supporting port operational planning, especially in areas with clear seasonal patterns in passenger flow. However, there is little literature that specifically focuses on Bakauheni Port in Lampung, which is one of the busiest ferry ports in Indonesia. Despite its strategic role, comprehensive forecasting studies for this port are still rare.

This study developed a SARIMA-based forecasting model to predict the number of ferry passengers at Bakauheni Port as a solution. By taking into account trends and seasonal patterns, the SARIMA model provides more reliable forecasts, helping port authorities to prepare ships better, optimize schedules, and improve service quality. This research aims to produce a ferry passenger number forecasting model and evaluate its accuracy to support operational planning at Bakauheni Port. The implementation of this model is expected to support data-driven decision making, improve service quality, and enhance operational efficiency at the port.

METHOD

Research Flow

This study combines various methods, such as literature analysis, which is useful for gaining an in-depth understanding and concepts relevant to the study. This is followed by the collection of relevant datasets taken from credible and reliable data sources. This is followed by structured and detailed stages through a flowchart. The flowchart can be seen in Figure 1.

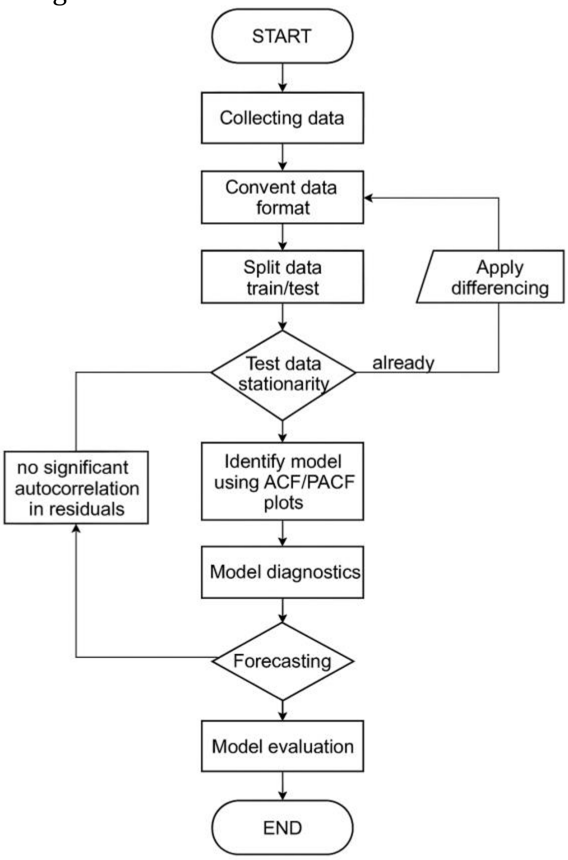

Figure 1. Flowchart of Research

Dataset

The dataset used in this study was sourced from the official website of Statistics Indonesia (BPS) for Lampung Province. The data were initially published in infographic format and subsequently transformed into tabular form using spreadsheet software to facilitate analysis. The dataset contains monthly records of ferry passenger volumes covering the period from 2014 to 2019.

Data Extraction

Data extraction is a process that involves taking visually presented information and transforming it into a more structured format. In this study, we used tabular data extracted from Infographic data on the comparison of ship passenger numbers from each month.

Split Data

Data splitting involves separating the dataset into two subsets with distinct roles in the modeling process. One subset is used to estimate and train the forecasting model, while the other is reserved for performance evaluation. This approach enables an unbiased assessment of the model's ability to generalize and produce reliable predictions on data that were not included during model estimation [26].

Stationarity Checking

Autocorrelation is a common issue in time series analysis and is closely linked to non-stationary behavior. Procedures used to transform a non-stationary series into a stationary form also function to reduce or eliminate autocorrelation, as stationarity inherently addresses such dependence [27]. A time series is considered stationary when its mean and variance remain constant or do not vary systematically over time [28]. The Augmented Dickey Fuller Test (ADF) is used in this study's stationarity testing. using the following generic equation.

$$\Delta Y_t = \beta_1 + \beta_2 + \delta Y_{t-1} + a_i \sum_{i=1}^{m} \Delta Y_{t-i} + \varepsilon_t \quad (1)$$

Where:
m: length of lag used

By conducting a stationary test using ADF, we can choose a model with only an intercept and a trend. In the Augmented Dickey-Fuller (ADF) test, if the p-value is less than 0.005, then the data is considered stationary. Here is the Model with an intercept only:

$$\Delta Y_t = \beta_1 + \delta Y_{t-1} + a_i \sum_{i=1}^{m} \Delta Y_{t-i} + \varepsilon_t \quad (2)$$

Model without intercept and trend

$$\Delta Y_t = \delta Y_{t-1} + a_i \sum_{i=1}^{m} \Delta Y_{t-i} + \varepsilon_t \quad (3)$$

Transformation will be carried out using the difference if the test results are declared non-stationary. Subtracting the Zt data from the Zt-1 data is the differencing procedure. The backward shift operator B is commonly employed in differencing procedures to address non-stationarity in the mean of a time series. When the series exhibits systematic changes over time, backward differencing is applied by computing the difference between successive observations, which helps suppress trends and other non-stationary components. The backshift operator shifts the observation index backward by one period, forming the basis for first-order differencing in time series transformation.

$$\beta Z_t = Z_{t-1} \quad (4)$$

Where:

B : differentiator (backward shift)

Z_t : Z value in period t

Z_{t-1} : Z value in period t-1

The notation paired with will shift the B Zt data 1 period back. For example, if a time series of data is not stationary, then the data can be made closer to stationary. Thus, the first differencing equation is as follows:

$$\Delta Z_t = Z_t - Z_{t-1} = Z_t - \beta Z_t = (1-\beta)Z_t \quad (5)$$

ACF (*Autocorrelation Function*)

The autocorrelation function (ACF) serves as a statistical measure for analyzing the relationship between observations in a time series at various lag lengths [29]. This analysis reveals how present values are associated with preceding observations, thereby indicating the degree of temporal dependence within the series [30], [31]. By examining these correlations, ACF facilitates the identification of structural patterns such as trends and seasonality and guides the determination of the appropriate order of the moving average component in ARIMA-based modeling frameworks[32], [33].

PACF (Partial *Autocorrelation Function*)

PACF is useful for the relationship between observed data and lag, while controlling for the effects of lag and the data being observed [34]. This relationship is very useful in identifying the order value (p) in the AutoRegressive model, a model that depends on its own error value in the previous data. PACF analyzes the significance value of the lag, which tells what values to include in the AutoRegressive.

Box-Cox Transformation

The Box-Cox transformation is used to reduce data skewness and stabilize variance, making the dataset more suitable for modeling [33]. This transformation is based on Tukey's earlier concept of data transformation and was further developed into a family of mathematical functions by Box and Cox in 1964, resulting in what is now known as the Box-Cox transformation:

$$y_t^{\lambda} = \left(y_t^{\lambda} - 1\right)/\lambda \qquad ;\text{For } \lambda \neq 0 \quad (6)$$

And

$$y_t^{\lambda} = log_e(y_i) \qquad ;\text{For } \lambda = 0 \quad (7)$$

The value λ can vary, which is useful in dealing with errors in the distribution. Each λ value will produce a different transformation, as follows.

Table 1. The Explanation of λ Value

No	λ Value	Explanation
1	**λ=1.00**	no transformation is applied; the data remain unchanged
2	**λ=0.50**	square root transformation
3	**λ = 0.33**	cube root transformation
4	**λ = 0.25**	fourth-root transformation
5	**λ = 0.00**	natural logarithm transformation
6	**λ = -0.50**	inverse square root transformation
7	**λ = -1.00**	reciprocal (inverse) transformation

AIC (Akaike Information Criterion)

The Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) are commonly employed to assess model performance by balancing goodness-of-fit against model complexity. Both measures incorporate penalties for an excessive number of parameters, thereby favoring models that provide an accurate yet parsimonious representation of the data. Models with smaller AIC or BIC values are typically considered more appropriate. The AIC statistic is computed using the following formula:

$$AIC = -2ln(L) + 2k \quad (8)$$

Where:
L: Likelihood of the model
k: Number of parameters

Models associated with smaller AIC values are generally considered preferable, as they reflect an appropriate compromise between explanatory accuracy and model simplicity. This criterion favors models that adequately capture the underlying data structure while avoiding overfitting or excessive parameter complexity.

SARIMA (Seasonal Autoregressive Integrated Moving Average)

The Seasonal Autoregressive Integrated Moving Average (SARIMA) framework is used to model and predict time series data characterized by recurring seasonal behavior. This approach combines seasonal and non-seasonal structures to capture systematic patterns such as long-term movements, cyclical dynamics, and periodic fluctuations in the data. The general formulation of the SARIMA model is stated as:

$$ARIMA(p,d,q)(P,D,Q)^S \quad (9)$$

The model components represented by lowercase parameters (p,d,q)(p, d, q)(p,d,q) correspond to the non-seasonal structure of the time series, capturing short-term dynamics and regular differencing effects. In contrast, the uppercase parameters (P,D,Q)(P, D, Q)(P,D,Q) describe the seasonal structure of the model, which accounts for periodic patterns and seasonal differencing present in the data. The notation s in the SARIMA equation indicates the seasonal period of the seasonal model. This satisfies the equation:

$$\phi_p(B^s)\phi_p(B)(1-B)^d(1-B^s)^D z_t = \theta_q(B)\theta_Q(B^S)\varepsilon_t \quad (10)$$

where:
B: backshift operator functions to shift data to the previous period
ε_t: error in period t

RESULTS AND DISCUSSION

The data used is the ferry passenger departure data in Bakauheni Lampung. The data used in this analysis were sourced from the Central Statistics Agency (BPS) and cover the period from January 2014 to December 2019. The values in the dataset are represented in thousands of people. The dataset has been split into training data, which accounts for 80%, and testing data, which makes up the remaining 20%. The data is presented in Table 2.

Table 2. Data on the Number of Ferry Passenger Departures in Bakauheni, Lampung

Year	January	February	...	December
2014	97.206	74.848	...	109.964
2015	106.496	6.497	...	155.793
2016	141.185	102.815	...	157.838
...	...	...	...	...
2019	116.240	84.745	...	140.370

Table 2 presents the monthly number of ferry passenger departures from Bakauheni Port, located in Lampung, Indonesia, spanning from the year 2014 to 2019. The data is categorized by year and month, with each cell representing the total number of ferry passengers (in thousands) who departed during that specific month. Only a few specific months are shown in the excerpt (January, February, and December), with other months represented by ellipses ("..."), indicating that the full data includes all months. January 2016 saw the highest January figure (141.185 or 141,185 passengers), while February 2015 had a significantly lower figure (6.497 or 6,497. In December, the number of passengers remained relatively high, suggesting a possible peak due to year-end travel.

Time Series Plot Identification

An initial visual inspection of the ferry passenger data is conducted to examine its temporal structure before model specification. This step provides a preliminary assessment of variability, recurring fluctuations, and seasonal behavior in the observed series.

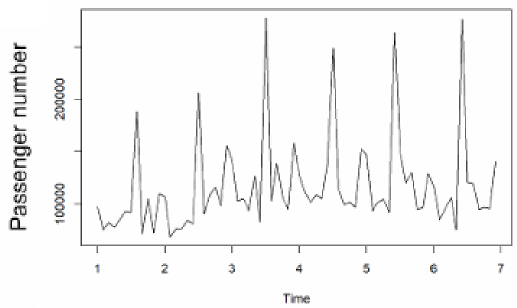

Figure 2. Passenger Departure Data Plot Through Bakauheni Port, Lampung

The data in Figure 2 plot shows seasonality, marked by repeated increases in certain periods. So, we can do forecasting using the Seasonal ARIMA (SARIMA) method. Figure 2 illustrates the time series plot of passenger departures through Bakauheni Port, Lampung, over a span of approximately seven years. The time is represented on the x-axis in terms of years, while the y-axis illustrates the number of passengers. From the plot, it is evident that the data exhibits a clear seasonal pattern, as indicated by the repeated spikes in passenger numbers at regular intervals—likely corresponding to annual events such as national holidays or religious celebrations like Eid al-Fitr [35]-[37]. These recurring peaks suggest that passenger traffic increases significantly during specific times of the year. Given this strong seasonality, the dataset is well-suited for forecasting using the Seasonal ARIMA (SARIMA) method, which is designed to model time series data with repeating seasonal trends [38]. Such insights are valuable for transportation authorities and ferry operators in planning for peak travel periods, ensuring adequate resources and smoother operations during times of high demand. Figure 3 explains the training and test data.

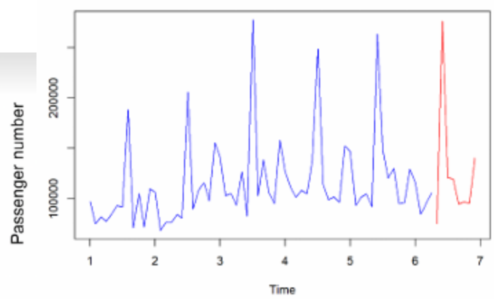

Figure 3. Dividing Train and Test Data

The next thing we do is to divide the training data and test data [19], [39], [40]. As shown in Figure 3, the data is divided into 80% for training, represented by a blue plot for modeling, and 20% for testing, indicated by a red plot, which is used to validate the model. The time series plot reveals that the data is non-stationary, indicating the need for a transformation using the Box-Cox method to address this issue.

Stationarity Test and Box-Cox Transformation

The time series data satisfy the stationarity requirements for reliable modeling following appropriate preprocessing. Variance stabilization is achieved through a Box–Cox transformation, while differencing is applied to remove trend and seasonal components. This procedure ensures a statistically robust foundation for subsequent model identification and estimation, as shown in Figure 4.

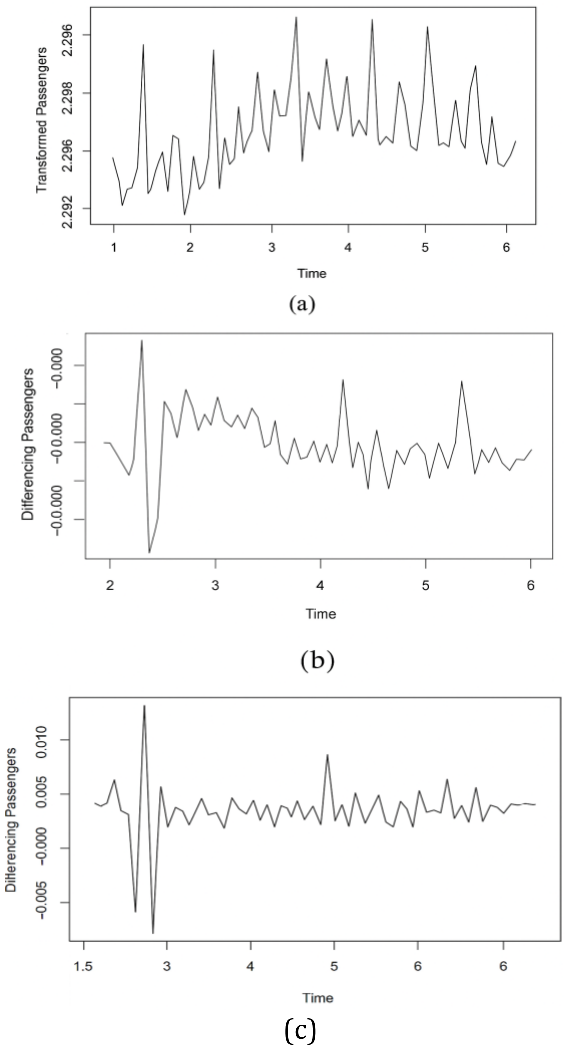

Figure 4. Box-Cox Transformation

a. Box-Cox Transformed Time Series Data
b. Seasonally Differentiated Time Series Data
c. Non-Seasonally Differentiated Time Series Data

The Box–Cox transformation was applied to stabilize variance and assess the stationarity of the time series [41]. Based on the Augmented Dickey–Fuller (ADF) test shown in Figure 4(a), the p-value of 0.358 indicates that the series was initially non-stationary. Seasonal differencing was then performed to remove recurring seasonal effects, followed by first-order differencing to eliminate the underlying trend, as illustrated in Figure 4(b). However, the ADF test still produced a p-value of 0.203, suggesting that stationarity had not yet been achieved. Consequently, additional non-seasonal differencing was applied, resulting in an ADF p-value of 0.01 in Figure 4(c), which confirms that the transformed series satisfies the stationarity assumption.

Model Identification

The data on the number of ferry passenger departures from Bakauheni, Lampung, exhibits seasonality, which is why differencing is performed using a lag of 12, corresponding to the annual period. The ACF and PACF plots after differencing are shown in Figure 5.

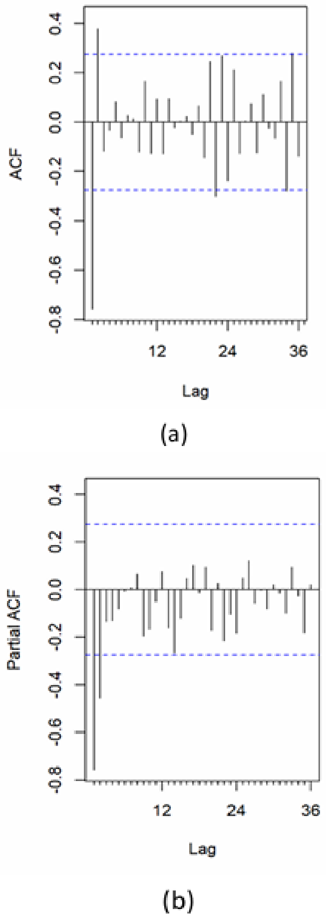

Figure 5. Plot of ACF and PACF After Differencing

Figure 5(a) illustrates that the ACF plot reveals autocorrelation between the values in the data at specific lags. In this plot, the ACF appears to cut the significant line (blue line) around the 1st lag and then disappears. This shows that the order of MA(q) may be at, $q = 1$. Based on Figure 5(b), it is known that the plot shows the PACF plot, PACF shows partial correlation between the larger lag values, after the influence of other lags disappears. The order of the AR (Autoregressive, p) model can be determined from the PACF pattern. As shown in Figure 5(b), the significance cutoff is observed at the first lag and partially at the following lag. This suggests that the AR(p) order could be either p=1 or p=2, depending on the degree of significance at each lag.

From observing the ACF and PACF graphs, several model assumptions were obtained [42], [43], [44]. Research by Dimri et al. [45] obtained ACF graphs that experienced discontinuity at certain lags, while PACF experienced discontinuity at the initial lag, indicating the existence of low-order Moving Average (MA) and Autoregressive (AR). This study proves that models with parameters determined by ACF and PACF produce smaller forecasting error values. By conducting a significance test on parameter estimates, the test is measured by the resulting p-value. If the coefficient is < 0.05, it is significant, but if the coefficient is > 0.05, it is not significant. Several models resulting from the identification are presented in Table 3.

Table 3. Model Identification

Model	Parameter	Coefficient	Note
$(2,1,1)(0,1,0)^{12}$	AR1	-0.7564	Significant
	AR2	-0.1958	Significant
	MA1	-0.4561	Significant
$(0,1,1)(0,1,0)^{12}$	MA1	-0.8985	Significant
$(1,1,0)(0,1,0)^{12}$	AR1	-0.7474	Significant
$(1,1,1)(0,1,0)^{12}$	AR1	-0.5643	Significant
	MA1	-0.6139	Significant
$(0,1,2)(0,1,1)^{12}$	MA1	-1.1560	Significant
	MA2	0.5114	Not Significant
$(1,1,2)(0,1,1)^{12}$	AR1	-0.4346	Significant
	MA1	-0.7758	Significant
	MA2	0.1847	Not Significant

The results indicate that most autoregressive (AR) and moving average (MA) parameters in the evaluated SARIMA specifications are statistically significant. In particular, the SARIMA $(2, 1, 1)(0, 1, 0)_{12}$ model exhibits fully significant parameters, suggesting that its structure is sufficient to capture the underlying data dynamics without requiring higher-order components. In contrast, the (0, 1, 2)(0, 1, 1)$_{12}$ and (1, 1, 2)(0, 1, 1)$_{12}$ models contain insignificant parameters, indicating that the inclusion of additional AR or MA terms does not meaningfully improve model performance. This finding is consistent with the study of Wang et al. [46], which reported that lower-order SARIMA models with statistically significant parameters tend to provide greater stability in forecasting seasonal passenger data.

Model selection is performed by evaluating competing specifications using the Akaike Information Criterion (AIC) together with forecasting error metrics. This approach enables the identification of a parsimonious model that balances goodness-of-fit and predictive performance. The calculated AIC, MAPE, and RMSE values are presented in Table 4. Based on the minimum error measures and the lowest AIC value, the SARIMA (2,1,1)(0,1,0)$_{12}$ model is selected as the most appropriate specification. Perform a diagnostic check on the best model that has been obtained. This is explained in Table 4.

Table 4. Summary of Error Values and AIC Values

Model	MAPE	RMSE	AIC
$(2,1,1)(0,1,0)^{12}$	0.0477168	0.0016637	-511.67
$(0,1,1)(0,1,0)^{12}$	0.05053	0.0018075	-502.39
$(1,1,0)(0,1,0)^{12}$	0.0515248	0.221839	-501.29
$(1,1,1)(0,1,0)^{12}$	0.04760	0.001670	-512.99
$(1,1,2)(0,1,0)^{12}$	0.047712	0.001665	-511.54

Table 4 provides a summary of the forecasting error metrics and Akaike Information Criterion (AIC) values obtained from the evaluated SARIMA models. Model performance is examined using Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and AIC, which collectively indicate predictive accuracy and model parsimony [47], [48]. Among the evaluated specifications, the SARIMA (2, 1, 1)(0, 1, 0)$_{12}$ model exhibits the strongest performance, as reflected by the lowest AIC value of −512.99, indicating the most appropriate balance between model fit and complexity. This model also produces a low RMSE value of 0.0016637, indicating strong predictive accuracy. Other candidate models exhibit higher AIC values, suggesting less parsimonious structures despite comparable error measures. These results confirm that the SARIMA (1, 1, 1)(0, 1, 0)$_{12}$ specification provides an optimal compromise between accuracy and complexity for forecasting seasonal ferry passenger departures at Bakauheni Port. This finding is consistent with Arit et al. [49] emphasizes that SARIMA model selection should prioritize minimal AIC values alongside low forecasting errors.

Model Diagnostics

The initial step in model diagnostics that can be done is to explore using the Q-Q Plot IN Figure 6.

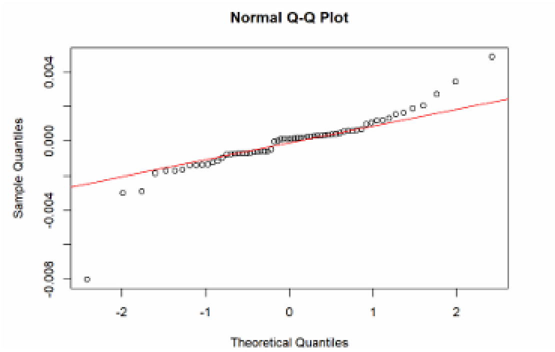

Figure 6. Normal QQ Plot

Figure 6 presents the Q–Q plot analysis, which indicates that the residuals are closely aligned with the normal reference line, suggesting that the residuals approximately follow a normal distribution. In addition, the residual plot reveals a random pattern with residuals scattered around zero; most residual points are around the diagonal line, with small deviations at the tail. This pattern indicates that the residual model approximates a normal distribution [50]. A summary of residual model test values is explained in Table 5.

Table 5. Summary of Residual Model Test Values

Testing Model	p-value
Ljung-Box	0.2556
Jarque bera	2.2e-16
Shapiro wilk	7.145e-06

Continuing with a formal test to examine the assumption of normality in the residual model, the Shapiro-Wilk test was performed, which can be seen in Table 5 with a p-value of 7.145e-06, indicating that the model has a normal residual distribution. Similarly, the Jarque-Bera test was performed, resulting in a p-value of 2.2e-16. These results indicate that the residuals follow a normal distribution.

Formal normality testing of the model residuals was conducted using the Shapiro–Wilk test, as reported in Table 5. The resulting p-value of 7.145 $\times 10^{-6}$ indicates that the residuals do not deviate substantially from the normality assumption. Consistent results were obtained from the Jarque–Bera test, which produced a p-value of 2.2 $\times 10^{-16}$, further supporting the normal distribution of residuals. Model adequacy was further evaluated through the Ljung–Box test to assess residual autocorrelation. A p-value greater than 0.05 indicates the absence of significant autocorrelation. The obtained p-value of 0.2556 exceeds this threshold, suggesting that the residuals are independently distributed. These diagnostic results confirm that the SARIMA $(2,1,1)(0,1,0)_{12}$ model satisfies key residual assumptions and is suitable for reliable forecasting.

This study, which applied the SARIMA (2, 1, 1)(0, 1, 0)12 model, produced a MAPE value of 11.47%, reflecting a relatively high level of forecasting accuracy. Although this accuracy is slightly lower than the findings of Multiningsih et al. [8], who reported a MAPE of 10.23% in their study at Semayang Port, the results are still comparable and show that the model works well in capturing passenger movement patterns. A study by [51] confirmed the existence of strong seasonal patterns in ferry passenger traffic, especially during holiday periods. In addition, visual analysis of the time series graph in this study showed clear annual seasons, which were in line with the patterns observed in the study at Yos Soedarso Port in Ambon. This consistency indicates that the SARIMA model is generally reliable for capturing seasonal fluctuations in ferry passenger data at various ports in Indonesia. The similarity of data patterns supports the use of SARIMA as a standard approach for forecasting in port operations [52].

Forecasting

Forecasting is performed to evaluate the predictive ability of the selected SARIMA model in capturing future passenger demand based on historical time series patterns. Model parameters are estimated, and future passenger volumes are projected to test how well the model represents underlying trends and seasonal fluctuations. This forecasting stage provides an empirical basis for assessing the model's suitability for short- to medium-term planning in ferry transport operations.

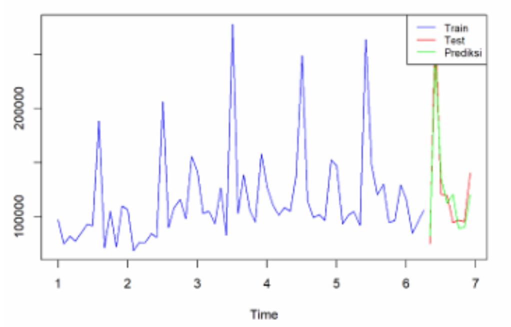

Figure 7. Forecasting Plot

Figure 7 illustrates the predicted passenger volumes for the following eight periods derived from the test dataset. In the figure, the blue line corresponds to the training series, the red line represents the observed test data, and the green line denotes the forecasted results. The predictive performance of the model is assessed using the Mean Absolute Percentage Error (MAPE), which measures the average percentage deviation between forecasts and actual values, together with an accuracy indicator that reflects the level of correspondence between predicted and observed data. Based on Figure 7, the MAPE is calculated to be 11.47%, which classifies the prediction as good, as it falls within the range of 10% ≤ MAPE < 20%, corresponding to an accuracy of 88.53%. The SARIMA model developed in this study demonstrates reliable performance in forecasting passenger numbers, maintaining an error rate within an acceptable range. Its robustness makes it suitable for short- to medium-term forecasting applications.

The results of this study offer important insights for transportation management and policy formulation. The application of the SARIMA approach demonstrates its capability to effectively represent both trend dynamics and seasonal patterns inherent in time-series data. This improves the accuracy of the model's forecasts and makes it applicable for practical planning in transportation systems such as ferry ports. This approach is also relatively easy to understand and computationally efficient, making it implementable by port authorities. This contributes to improved operational efficiency and passenger satisfaction, while also laying the foundation for future research that can integrate more complex variables or more sophisticated models.

The availability of data-driven forecasting models allows for more accurate vessel scheduling, more efficient allocation of operational resources, and reduced losses due to passenger and vehicle delays and congestion, thereby improving overall economic efficiency. More precise predictions of passenger and vehicle flows enable port management to proactively respond to surges in demand, expedite service processes, and reduce waiting times, which in turn improves user satisfaction and strengthens the port's image as a major ferry transportation hub connecting Sumatra and Java. Beyond operational benefits, optimized ferry service scheduling and congestion mitigation also contribute to environmental sustainability by reducing fuel consumption and emissions from maritime transport activities, aligning port operations with the Sustainable Development Goals (SDGs), particularly those related to clean energy and resilient infrastructure. This research provides a meaningful foundation for improving the long-term efficiency, service quality, and sustainability of the ferry transportation system.

LIMITATION

This study also has several limitations that need to be considered, including the SARIMA model developed using only historical data without incorporating external factors such as weather conditions, economic variables, or special events, which can also affect passenger numbers. In addition, model performance may decline over longer forecasting horizons due to the accumulation of uncertainty, so long-term forecasting results need to be interpreted with caution.

CONCLUSION

The SARIMA $(2,1,1)(0,1,0)^{12}$ model was identified as the best model for forecasting ferry passenger departures from 2014 to 2019. It effectively captures strong seasonal patterns and demonstrates good predictive performance, with a MAPE of 11.47% and an accuracy of 88.53%. With low AIC values, this model is suitable for supporting port operational planning by helping anticipate passenger fluctuations and optimize services. This study successfully developed a SARIMA model that accurately predicts ferry passenger numbers, aiding planning and operations at Bakauheni Port. While the model has limitations, such as not accounting for external factors, its results provide a solid foundation for data-driven decision-making and service improvement in the future.

AUTHOR CONTRIBUTION

K.M.A. and L.M. designed the research concept and design, curated the ferry passenger time series dataset, developed and modeled SARIMA, and prepared the manuscript (lead author and corresponding author). E.S. and R.N. contributed to data preprocessing, Box-Cox transformation, stationarity testing (ADF), and model identification through ACF and PACF. P. and G.A.K. conducted a literature review, analysis of related research, and preparation of the theoretical basis and research methodology. A.M. and I.S. played a role in validating the forecasting results, evaluating model performance using MAPE, RMSE, and AIC, analyzing residual diagnostics, and editing and adjusting the manuscript format.

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REFERENCES

[1] S. Fatimah, *Pengantar Transportasi*. Ponorogo: Myria, 2019.

[2] S. del Saz-Salazar and B. Tovar, “On ferry users’ willingness to pay for improving environmental quality: A case study for the Canary Islands,” *Transport Policy*, vol. 154, no. 1, pp. 61–72, 2024, doi: 10.1016/j.tranpol.2024.05.023.

[3] BPS Lampung, “Perkembangan Transportasi Provinsi Lampung April 2023,” *https://lampung.bps.go.id/id/pressrelease/2023/06/05/1117/perkembangan-transportasi-provinsi-lampung-april-2023.html*. 2023. [Online]. (Accessed March. 22, 2025).

[4] M. Down, Z. Baldock, D. Picknoll, and C. Bulsara, “Seeking consensus : An e - delphi study exploring the concept of success in australian outdoor education,” *Journal of*

Outdoor and Environmental Education, vol. 1, no. 1, pp. 1-19, 2025, doi: 10.1007/s42322-025-00206-7.

[5] M. Silfiani and F. N. Hayati, "Forecasting number of train passengers using time series regression integrated calendar variation and covid 19 intervention," *Jurnal Matematika Sains dan Teknologi*, vol. 25, no. 1, pp. 09–22, 2024.

[6] O. Hannonen, T. Aguiar Quintana, and X. Y. Lehto, "A supplier side view of digital nomadism: The case of destination Gran Canaria," *Tourism Management*, vol. 97, no. 8, pp. 1-15, 2023, doi: 10.1016/j.tourman.2023.104744.

[7] J. C. Centeno *et al.*, "Evaluation of factors contributing to traffic flow along jose abad," *International Journal Of Progressive Research In Science And Engineering*, vol. 4, no. 06, pp. 470–477, 2023.

[8] M. Multiningsih, E. Siswanah, and M. Saleh, "Forecasting the number of ship passengers with SARIMA approach (a case study: Semayang port, Balikpapan city)," *JTAM (Jurnal Teori dan Aplikasi Matematika)*, vol. 6, no. 4, pp. 1-21, 2022, doi: 10.31764/jtam.v6i4.10211.

[9] K. Trepte and J. B. Rice, "An initial exploration of port capacity bottlenecks in the USA port system and the implications on resilience," *International Journal of Shipping and Transport Logistics*, vol. 6, no. 3, pp. 339–355, 2014, doi: 10.1504/IJSTL.2014.060800.

[10] L. Manzanero and K. Krupp, "Forecasting container freight rates – an econometric analysis," Dept. Finance and Strategic Management, Copenhagen Business School, Copenhagen, MM, Denmark, 2009.

[11] F. Parola, G. Satta, T. Notteboom, and L. Persico, "Revisiting traffic forecasting by port authorities in the context of port planning and development," *Maritime Economics & Logistics*, vol. 23, no. 3, pp. 1-51, 2020, doi: 10.1057/s41278-020-00170-7.

[12] T. N. Cuong, H.-S. Kim, and S.-S. You, "Data analytics and throughput forecasting in port management systems against disruptions: A case study of Busan Port," *Maritime Economics & Logistics*, vol. 25, no. 1, pp. 1-29, 2022, doi: 10.1057/s41278-022-00247-5.

[13] Nasution, *Manajemen Transportasi*. Bogor: Ghalia Indonesia, 2015.

[14] I. P. Y. Prabhadika, N. K. T. Tastrawati, and L. P. I. Harini, "Peramalan persediaan infus menggunakan metode autoregressive integrated moving average (arima) pada rumah sakit umum pusat sanglah," *Jurnal Matematika*, vol. 7, no. 2, pp. 129–133, 2018, doi: 10.24843/MTK.2018.v07.i02.p194.

[15] M. M. Kasawehi, D. Hatidja, and Y. A. R. Langi, "Prediction the number of ship passengers in ports tagulandang using the triple exponential smoothing (TES) method," *Jurnal Ilmiah Sains*, vol. 23, no. 2, pp. 108–117, 2023, doi: 10.35799/jis.v23i2.48886.

[16] D. C. Montgomery, C. L. Jennings, and M. Kulahci, *Introduction to Time Series Analysis and Forecasting*. Canada: John Wiley & Sons, 2007.

[17] W. Butarbutar, M. S. N. Van Delsen, and R. J. Djami, "Application of the seasonal autoregressive integrated moving average (SARIMA) method to forecast the number of vessel passenger departures at yos soedarso ambon port," vol. 4, no. 2, pp. 105–118, 2023.

[18] W. Reza, "Comparison of the accuracy of seasonal data prediction values using SARIMA and winter exponential smoothing on the number of ship passengers in Batam City," *Jurnal Pijar Mipa*, vol. 18, no. 4, pp. 632–637, 2023, doi: 10.29303/jpm.v18i4.5469.

[19] D. W. Saputro *et al.*, "Predicting the number of train passengers in java island using SARIMA model," *EKSAKTA: Journal of Sciences and Data Analysis*, vol. 4, no. 2, pp. 40–48, 2023, doi: 10.20885/EKSAKTA.vol4.iss2.art5.

[20] S. E. Sim, K. G. Tay, A. Huong, and W. K. Tiong, "Forecasting electricity consumption using SARIMA method in IBM SPSS software," *Universal Journal of Electrical and Electronic Engineering*, vol. 6, no. 5, pp. 103–114, 2019, doi: 10.13189/ujeee.2019.061614.

[21] S. Kumaria and P. Muthulakshmi, "SARIMA model: An efficient machine learning technique for weather forecasting," *Procedia Computer Science*, vol. 235, no. 1, pp. 656–670, 2023, doi: 10.1016/j.procs.2024.04.064.

[22] S. S. Ishak *et al.*, “Indonesian consumer price index forecasting using autoregressive integrated moving average,” *International Journal of Electronics and Communications Systems*, vol. 3, no. 1, pp. 33-40, 2023, doi: 10.24042/ijecs.v3i1.18252.

[23] R. Yulianti, N. T. Amanda, K. A. Notodiputro, Y. Angraini, and L. N. A. Mualifah, “Comparison of SARIMA and sarimax methods for forecasting harvested dry grain prices in indonesia,” *Barekeng*, vol. 19, no. 1, pp. 319–330, 2025, doi: 10.30598/barekengvol19iss1pp319-330.

[24] M. F. Zahid, A. Fitrianto, P. Silvianti, and A. Alamudi, "Comparison between SARIMA and DeepAR with optuna hyperparameter optimization for estimating rice production data in indonesia", *Indonesian Journal of Statistics & Its Applications,* vol. 8, no. 2, pp. 1-15, 2024, doi: 10.29244/ijsa.v8i2p95-111.

[25] C. Li, “Comparison of forecasting eff ect of SARIMA model and holt-winters smoothing based on GDP,” *Universe Scientific Publishing*, vol. 6, no. 2, pp. 71–81, 2021.

[26] L. F. Tokan and A. Hermawan, “Implementasi model SARIMA untuk memprediksi produksi minyak kelapa sawit,” *Jurnal Fasilkom*, vol. 13, no. 3, pp. 456–463, 2023, doi: 10.37859/jf.v13i3.6033.

[27] P. Utomo and A. Fanani, “Peramalan jumlah penumpang kereta api di indonesia menggunakan metode seasonal autoregressive integrated moving average (SARIMA),” *Jurnal Mahasiswa Matematika ALGEBRA*, vol. 1, no. 1, pp. 169–178, 2020.

[28] J. W. Osborne, “Improving your data transformations: Applying the Box-Cox transformation,” *Practical Assessment, Research and Evaluation*, vol. 15, no. 12, pp. 1-9, 2010, doi: 10.7275/qbpc-gk17.

[29] G. M. Tinungki, “The analysis of partial autocorrelation function in predicting maximum wind speed,” *IOP Conference Series: Earth and Environmental Science*, vol. 235, no. 1, pp. 1-13, 2019, doi: 10.1088/1755-1315/235/1/012097.

[30] L. T. Kreutzer *et al.*, “S-ACF : A selective estimator for the autocorrelation function of irregularly sampled time series,” *Monthly Notices of the Royal Astronomical Society,* vol. 522, no. 4, pp. 5049–5061, 2023, doi: 10.1093/mnras/stad1223.

[31] H. Lütkepohl and M. Kratzig, *Applied Times Series*. Cambridge: University Press, 2004.

[32] I. Syahrini, R. Radhiah, W. F. Damanik, N. Sciences, U. S. Kuala, and B. Aceh, “Application of the autoregressive integrated moving average (ARIMA) box-jenkins method in forecasting inflation rate in aceh,” *Transcendent Journal of Mathematics and Application ISSN*, vol. 2, no. 1, pp. 27–33, 2023.

[33] Y. Ensafi, S. H. Amin, G. Zhang, and B. Shah, “Time-series forecasting of seasonal items sales using machine learning – A comparative analysis,” *International Journal of Information Management Data Insights*, vol. 2, no. 1, pp. 1-16, 2022, doi: 10.1016/j.jjimei.2022.100058.

[34] H. Farhangi, “The path of the smart grid,” *IEEE Power and Energy Magazine*, vol. 8, no. 1, pp. 18–28, 2010, doi: 10.1109/MPE.2009.934876.

[35] M. S. Akbar, D. A. Safira, R. Fadhilah, S. Damayanti, Riskianto, and K. Puthy, “Eid al-fitr influences the number of train passengers on the Sumatra Island (calendar variations time series model),” *Barekeng*, vol. 18, no. 4, pp. 2191–2202, 2024, doi: 10.30598/barekengvol18iss4pp2191-2202.

[36] F. Dobruszkes, J. M. Decroly, and P. Suau-Sanchez, “The monthly rhythms of aviation: a global analysis of passenger air service seasonality,” *Transportation Research Interdisciplinary Perspectives*, vol. 14, no. 6, pp. 1-16, 2022, doi: 10.1016/j.trip.2022.100582.

[37] E. Panzera, T. de Graaff, and H. L. F. de Groot, “European cultural heritage and tourism flows: The magnetic role of superstar world heritage sites,” *Papers in Regional Science*, vol. 100, no. 1, pp. 101–122, 2021, doi: 10.1111/pirs.12562.

[38] X. Liu, Z. Lin, and Z. Feng, “Short-term offshore wind speed forecast by seasonal ARIMA - A comparison against GRU and LSTM,” *Energy*, vol. 227, no. 7, pp. 1-24, 2021, doi: 10.1016/j.energy.2021.120492.

[39] S. Y. Ilu and R. Prasad, "Improved autoregressive integrated moving average model for COVID-19 prediction by using statistical significance and clustering techniques," *Heliyon*, vol. 9, no. 2, pp. 1-7, 2023, doi: 10.1016/j.heliyon.2023.e13483.

[40] F. M. Khan and R. Gupta, "ARIMA and NAR based prediction model for time series analysis of COVID-19 cases in India," *Journal of Safety Science and Resilience*, vol. 1, no. 1, pp. 12–18, 2020, doi: 10.1016/j.jnlssr.2020.06.007.

[41] V. M. Guerrero and R. Perera, "Variance stabilizing power transformation for time series," *Journal of Modern Applied Statistical Methods*, vol. 3, no. 2, pp. 357–369, 2004, doi: 10.22237/jmasm/1099267740.

[42] I. Syahzaqi, Sediono, A. C. Anggakusuma, E. E. Wieldyanisa, and S. S. Oktavia, "Modeling and forecasting the total volume of goods transported by rail in indonesia using seasonal autoregressive integrated moving average (SARIMA)," *Barekeng*, vol. 19, no. 2, pp. 829–842, 2025, doi: 10.30598/barekengvol19iss2pp829-842.

[43] M. Balawi and G. Tenekeci, "Time series traffic collision analysis of london hotspots: Patterns, predictions and prevention strategies," *Heliyon*, vol. 10, no. 4, pp. 1-29, 2024, doi: 10.1016/j.heliyon.2024.e25710.

[44] I. U. Moffat and E. A. Akpan, "White Noise analysis: A measure of time series model adequacy," *Applied Mathematics*, vol. 10, no. 11, pp. 989–1003, 2019, doi: 10.4236/am.2019.1011069.

[45] T. Dimri, S. Ahmad, and M. Sharif, "Time series analysis of climate variables using seasonal ARIMA approach," *Journal of Earth System Science*, vol. 129, no. 1, pp. 1-16, 2020, doi: 10.1007/s12040-020-01408-x.

[46] Y. Wang, R. Jia, F. Dai, and Y. Ye, "Traffic flow prediction method based on seasonal characteristics and SARIMA-NAR model," *Applied Sciences*, vol. 12, no. 4, pp. 1-18, 2022, doi: 10.3390/app12042190.

[47] S. Saigal and D. Mehrotra, "Performance comparison of time series data using predictive data mining techniques," *Advances in Information Mining*, vol. 4, no. 1, pp. 57–66, 2012.

[48] L. Zhou, P. Zhao, D. Wu, C. Cheng, and H. Huang, "Time series model for forecasting the number of new admission inpatients," *BMC medical informatics and decision making*, vol. 18, no. 1, pp. 1-11, 2018, doi: 10.1186/s12911-018-0616-8.

[49] J. Arlt and P. Trcka, "Automatic SARIMA modeling and forecast accuracy," *Communications in Statistics-Simulation and Computation*, vol. 50, no. 10, pp. 2949–2970, 2021, doi: 10.1080/03610918.2019.1618471.

[50] G. E. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time series analysis: forecasting and control*. Canada: John Wiley & Sons, 2015.

[51] J. Liu, W. Shi, and P. Chen, "Exploring travel patterns during the holiday season—A case study of Shenzhen Metro system during the Chinese Spring festival," *ISPRS International Journal of Geo-Information*, vol. 9, no. 11, pp. 1-22, 2020, doi: 10.3390/ijgi9110651.

[52] J. Farhan and G. P. Ong, "Forecasting seasonal container throughput at international ports using SARIMA models," *Maritime Economics & Logistics*, vol. 20, no. 1, pp. 131–148, 2018, doi: 10.1057/mel.2016.13.